CL141

ENGINEERING MECHANICS

Question Bank

First Year B.Tech. (CL/ME/EE)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty Of Technology & Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

Academic year: 2018-19

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

M S Patel Department of Civil Engineering

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CH: 2.1-COPLANAR CONCURRENT FORCE SYSTEM

I) LAW OF PARALLELOGRAM AND TRIANGULAR LAW

1) The resultant of two forces P and Q is at right angle to P. Show that the angle between the forces is $\cos^{-1}(-P/Q)$.
2) The resultant of two forces P and Q is R. If Q is doubled, the new resultant is perpendicular to P. Show that Q=R.
3) Find the angle between two equal forces of magnitude P when their resultant is 1.5 P. What will be the maximum value of resultant?

   (Ans : θ =82.82°,R=2P)
4) The angle between two forces is 120°, If their resultant makes an angle of 70° with the smaller force whose magnitude is 10 kN, calculate the magnitude of the larger force and also that of their resultant.

   (Ans : R=12.27 kN , R=11.31 kN)
5) The sum of two concurrent forces P and Q is 270 N and their resultant is 180 N. The angle between the force P and resultant R is 90°. Find the magnitude of each force and angle between them.

   (ANS : Q =195 N, P =75 N, θ=112.58°)
6) The force F1 is of 100N magnitude. The resultant of the force F1 and unknown force F2 is 100N in magnitude. Find the magnitude of F2 if resultant is perpendicular to F1.

   (Ans : F2= 141.42N, $\theta = 135^0$)
7) Two compressive forces of 20 kN and 30 kN are acting at a point with an angle of 60° between them. Calculate magnitude and direction of the resultant force using law of triangle of forces.

   (Ans : R= 43.58N, $\alpha = 36.58^0$ w.r.t 20 kN force)
8) Find the resultant of two forces, P and P/2 acting on a body with angle between them 135°. Use law of triangle of forces.

   (Ans : R= 0.74P, $\alpha = 28.68^0$)

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CH: 2.1-COPLANAR CONCURRENT FORCE SYSTEM

II) ORTHOGONAL AND NON ORTHOGONAL COMPONENT

1) Resolve 75 N force as shown in **figure (1)** along mm and nn axes.
   (Ans : 54.90 N & 67.24 N)
2) Resolve 100 N force acting on triangular lamina as shown in **figure (2)** along AB and parallel to BC.
   (**Ans: 115.47 N & 57.73 N)**
3) Resolve 150N force parallel and perpendicular to axis a-a shown in **Figure (3)**. **(Ans: 35.627N , 145.707 N)**
4) Resolve 5000N force into two non-orthogonal components such that one component is along m-m axis and other component is along n-n axis. Refer **figure (4)**
   (Ans: 2231.6 N & 3481.29 N)
5) A force is acting at an angle of 60 °. Find the components along the axes a-a and b-b as shown in **figure (5).**
   (Ans: along a-a = 50kN & b-b = 50 kN)

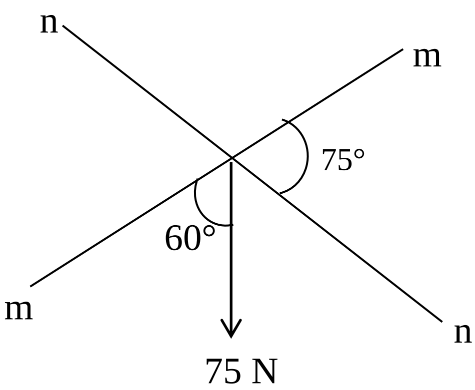

Figure 1

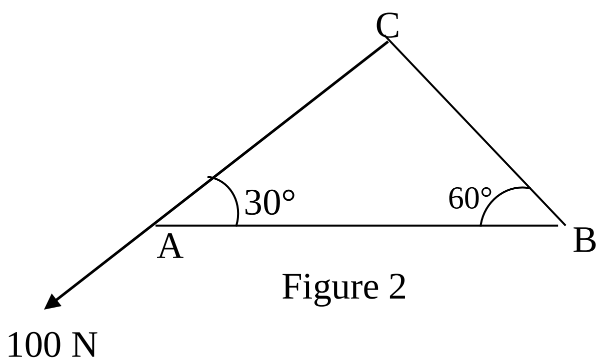

Figure 2

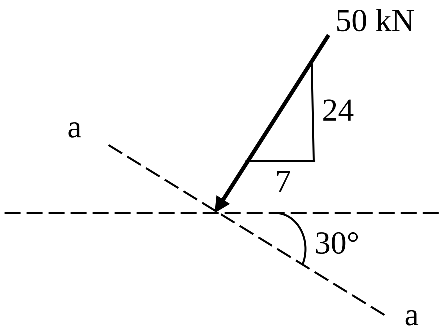

Figure 3

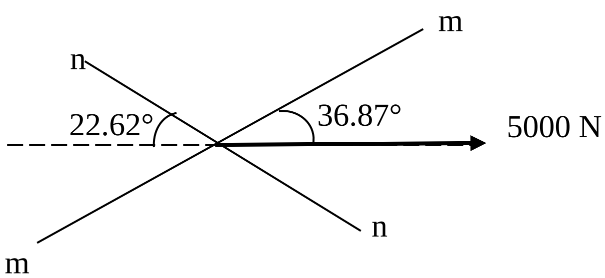

Figure 4

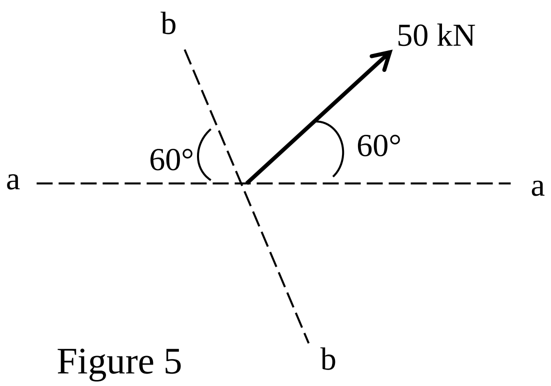

Figure 5

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CH: 2.1-COPLANAR CONCURRENT FORCE SYSTEM

III) METHOD OF RESOLUTION

1) A system of four forces shown in **figure (1)** has resultant 50 N along +ve X axis. Determine magnitude and direction of unknown force P.

   (Ans: P=100 N, θ = 60°)

2) The resultant of three forces shown in **figure (2)** is along X axis. Determine angle α and magnitude of resultant.

   (Ans:R=75.23 N, α=-62°)

3) Determine magnitude and direction of resultant force for the system shown in **figure (3)**.

   (Ans:R=201.00 N, α=-41°)

4) Find equilibrant of the force system shown in **figure (4).**

   (Ans:R=71.71 N, α=-80.80°)

5) Calculate the magnitude and direction of resultant of coplanar concurrent forces shown in **figure (5).** If you consider resultant as one force then what is the resultant of R, P_1, P_2, P_3?

   (Ans:R=311.46 N, α=-27.56°, R=573.12 N)

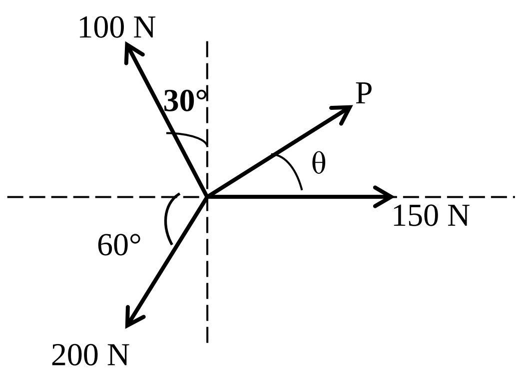

Figure 1

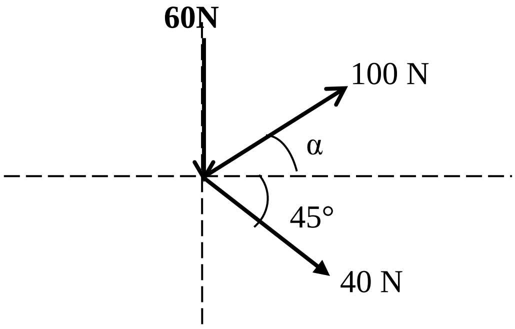

Figure 2

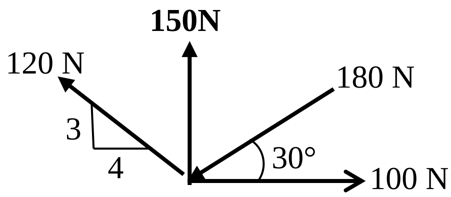

Figure 3

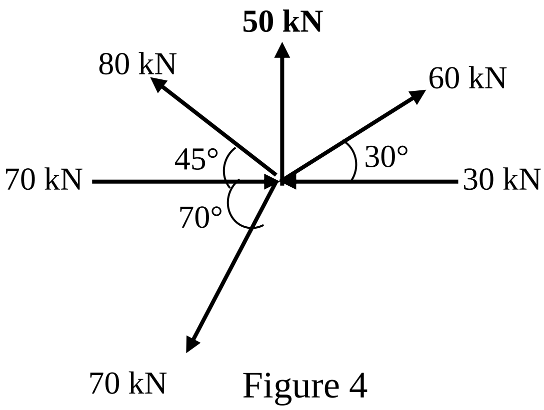

Figure 4

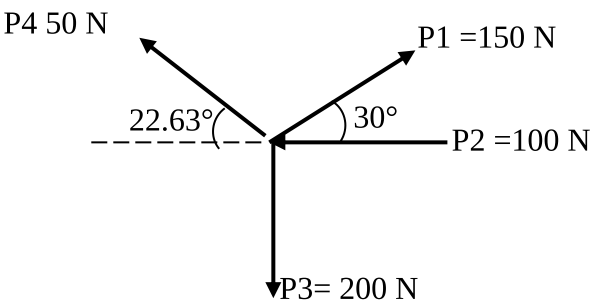

Figure 5

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CH: 2.2- MOMENTS AND COUPLES

I) MOMENT

1) Determine the moment of the force with respect to given point A for each of the system shown in **figure (1).**

   (Ans: (a) MA = -15.59 N.m, (b) MA = 56 N.m, (c) MA= 15.6 N.m)

2) Determine the moment of 6 kN force shown in **figure (2)** about point A by resolving the force into horizontal and vertical components. Also find out the perpendicular distance of point A from the line of action of 6 kN force by principle of moments.

   (Ans:MA = 0.624 kN.m, d = 0.104 m)

3) A force of 500 N passing through A (3, 0, 5 m) and B (4, 4, 6 m). Determine its moment about point origin O and C (3, 2, -1 m).

   (Ans:MO = 2758.8 Nm, MC= 3153.6 Nm)

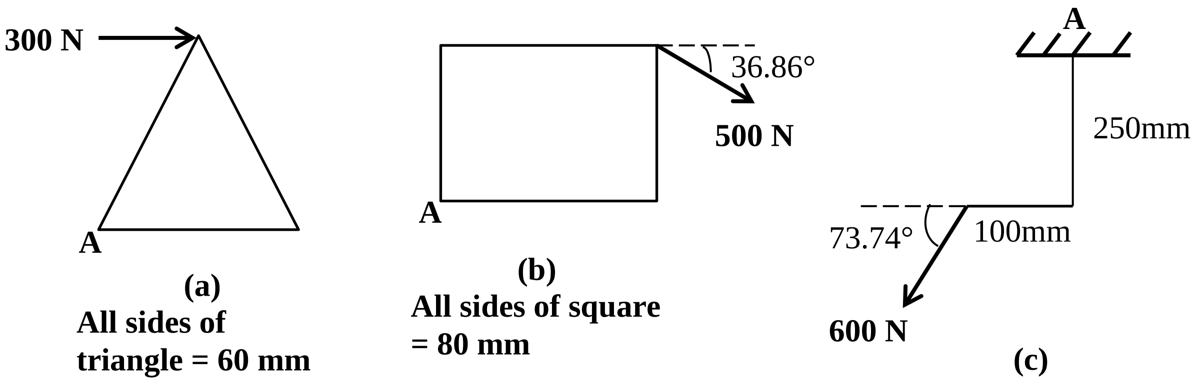

Figure 1

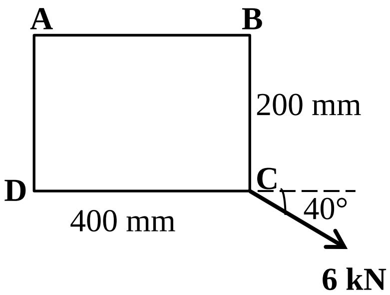

Figure 2

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CH: 2.2- MOMENTS AND COUPLES

II) EQUIVALENT COUPLE AND EQUIVALENT FORCE SYSTEM

1) Determine the moment of a couple with respect to given point E and F for the system shown in **figure (1).** What is your observation?

   (Ans: ME = 24 kN.m, MF = 24 kN.m)

2) Do the couples shown in **figure 2(a)** and **figure 2(b)** are equivalent?

3) Determine a single force and its distance from A which is equivalent to the force system shown in **figure 3**.

   (Ans:MA = 1.8 kNm, d = 0.3 m to the right of A)

4) Replace 200 N force by an equivalent force couple system about C for the force system shown in **figure 4**.

   (Ans:MC = 7464 N.cm)

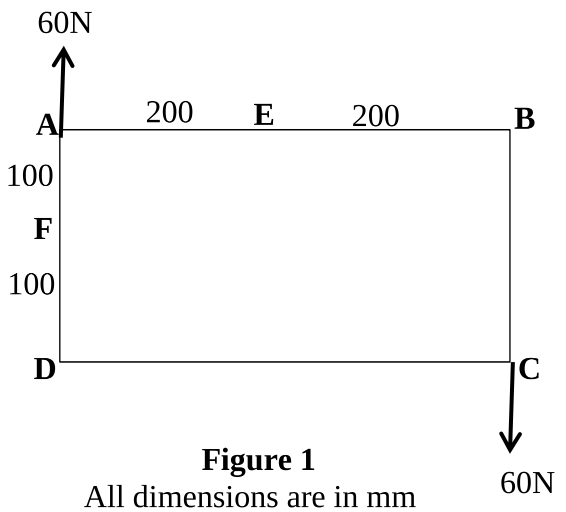

Figure 1
All dimensions are in mm

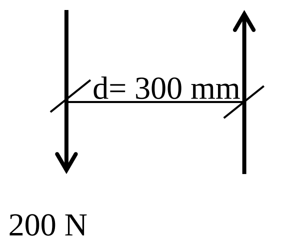

Figure 2(a)

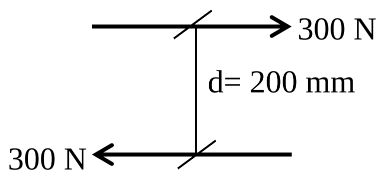

Figure 2(b)

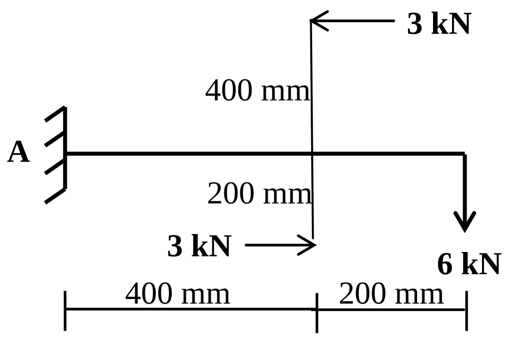

Figure 3

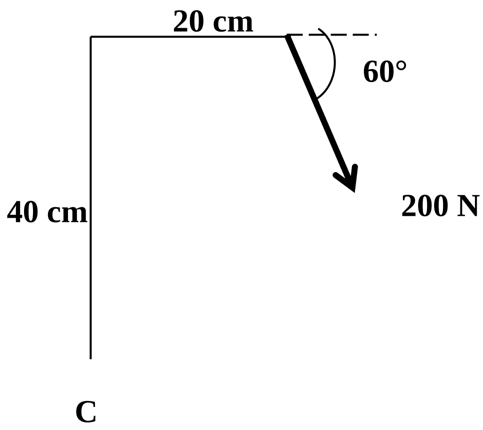

Figure 4

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CH: 2.3- COPLANAR NON-CONCURRENT FORCE SYSTEM

III) RESULTANT OF COPLANAR NON CONCURRENT FORCE SYSTEM

1) Find magnitude and direction of resultant with respect to O for a system shown in **figure (1).**

   (Ans: R=35.34 kN, α= 75, X=1.923m from O)

2) A coplanar non concurrent system of force is shown in **figure (2).** Determine the magnitude, direction and position of the resultant of the system with respect to A.

   (Ans: R=2013.95 N, θ = 59.51° , y = 1.068 m from A on AB)

3) Determine the resultant of the four coplanar non concurrent forces shown in **figure (3)** with respect to B.

   (Ans: R28.28 N, X=2.83 m from B)

4) Find magnitude, direction and location of resultant of force system shown in **figure (4)** with respect to 'O'.

   (Ans: R= 46.91 kN, x = 4.92 units)

5) A square of side 1m is acted upon by the force system as shown in **figure 5**. Find the resultant of the system in terms of magnitude; position and direction with respect to point A.

   (Ans: R= 28.28 N, x = 0.354 m from A)

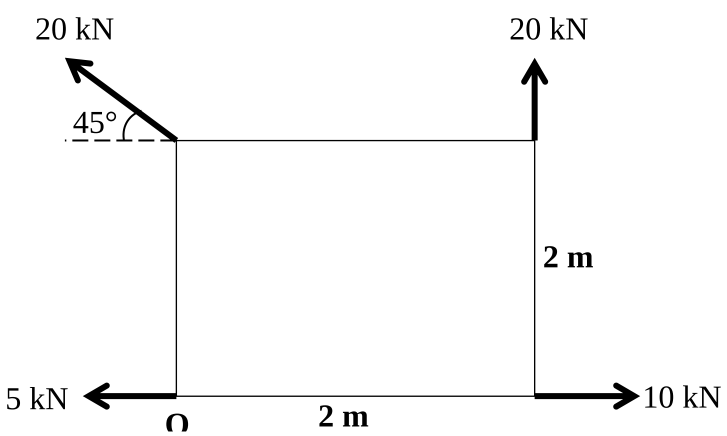

Figure 1

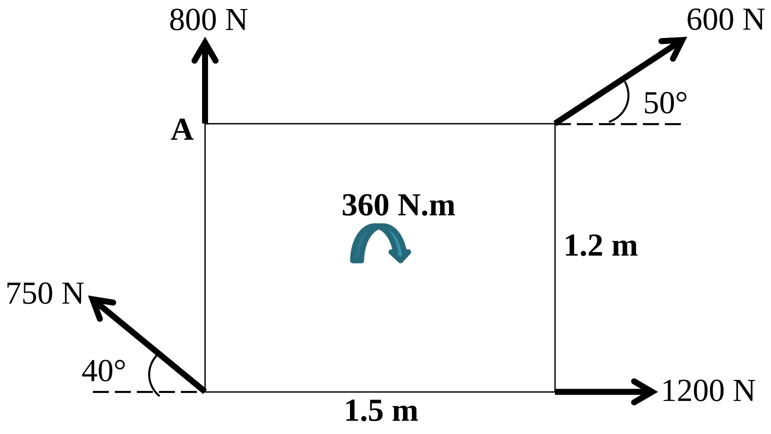

Figure 2

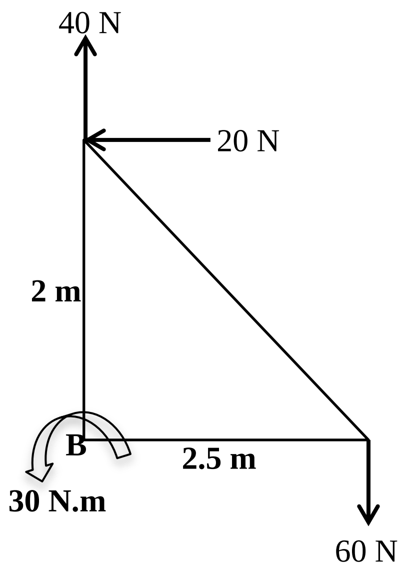

Figure 3

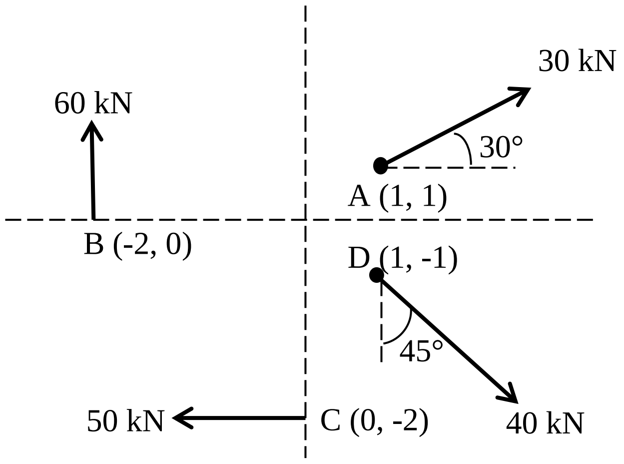

Figure 4

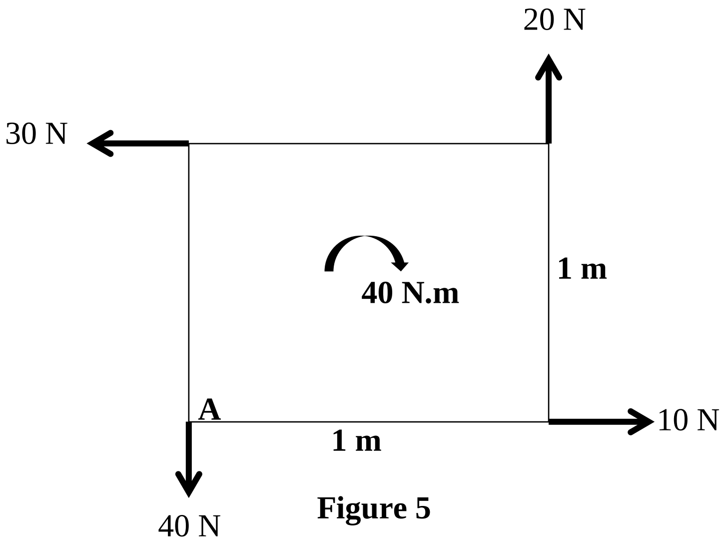

Figure 5

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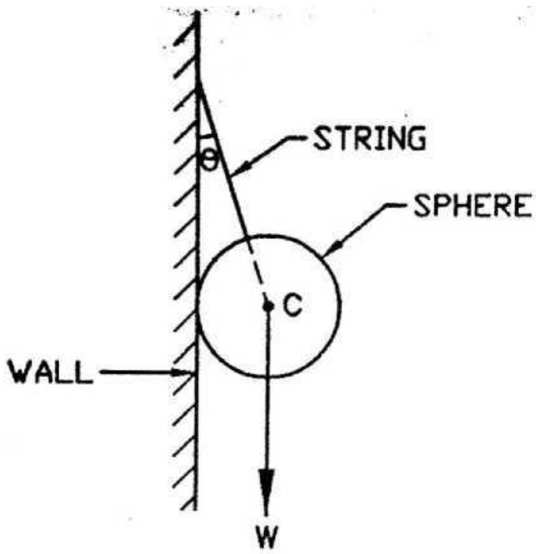
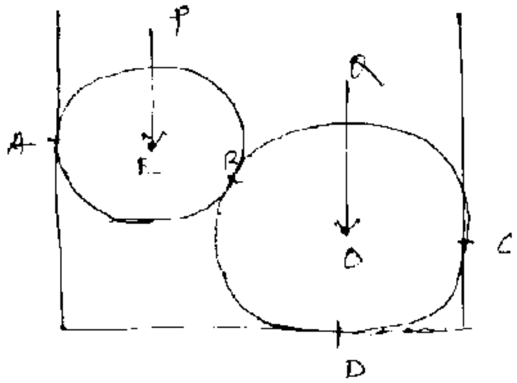
TYPE: 1 FREE BODY DIAGRAM

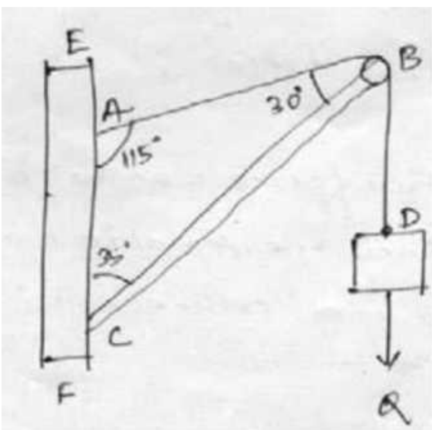
1. Draw the free body diagram of the body, at the joint C.



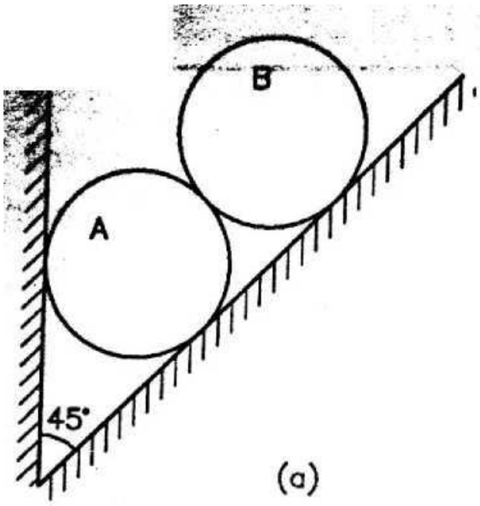
2. Two spheres of weight P and Q rest inside a hollow cylinder which is resting on a horizontal force. Draw the free body diagram of both the spheres separately.



3. Draw the free body diagram of the figure shown below with angle.

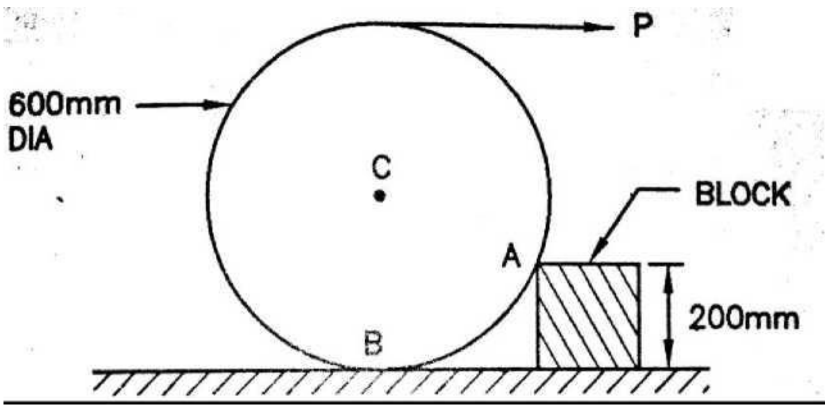


4. Two rollers of weight P and Q are supported by an inclined plane and vertical walls as shown in the figure. Draw the free body diagram of both the rollers separately.



(a)

5. Draw the free body diagrams of the following figures.



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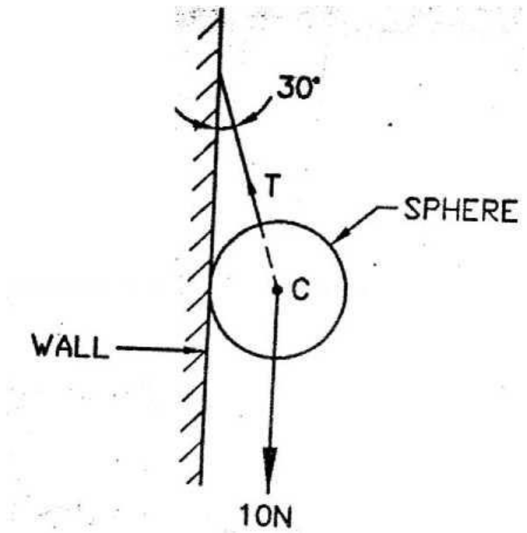
TYPE: 2 LAMI'S THEOREM AND TENSION IN STRING

1. State and explain Lami's Theorem.
2. A sphere weighing 10N is hanged as shown in fig. Find tension in the rope and reaction of wall.

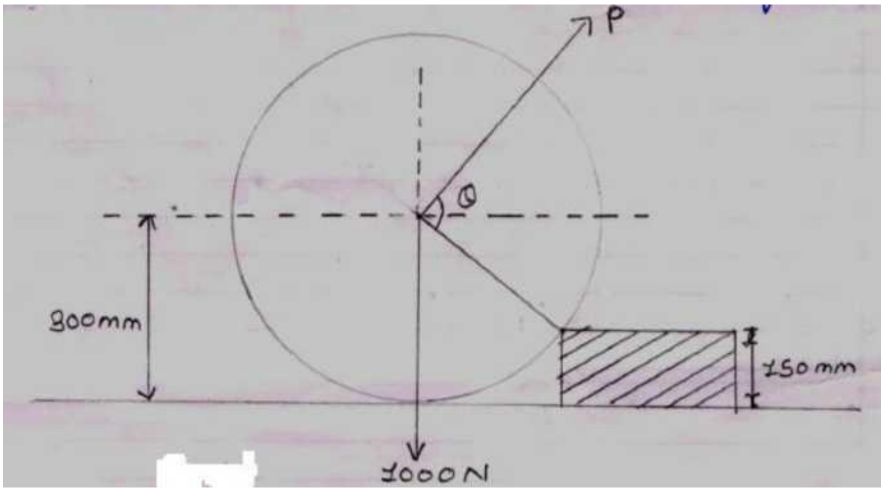
Ans. R_w=5.77N, T_r=11.55N



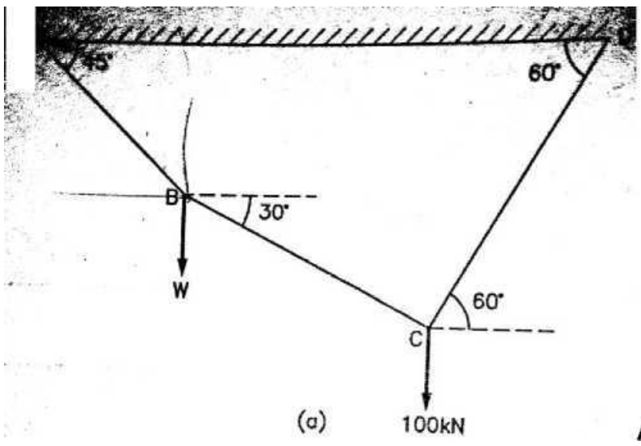
3. A load of 100kN is hung by means of a rope attached to a hook in horizontal ceiling. What horizontal force should be applied so that rope makes 60^0 with the ceiling? Draw free body diagram and calculate force and tension in rope. Also calculate minimum force required if angle of rope with ceiling remains 60^0.

Ans. 1) P1=57.74kN, T1=115.47kN **2)** P2=50kN, T2= 86.60kN

4. A uniform wheel of 600mm diameter and weight 1000N rests against rectangular obstacle which when acting through the least force required which when acting through the center of the wheel will just turn the wheel over the corner of the block. **(Ans. P=866N, R= 500N)**



5. A weight 'W' is suspended from the ceiling with the help of rope as shown in figure. Find the value of weight 'W'.

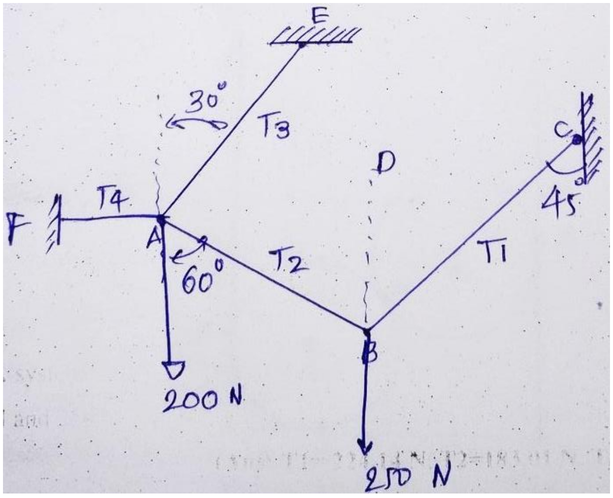


(a)

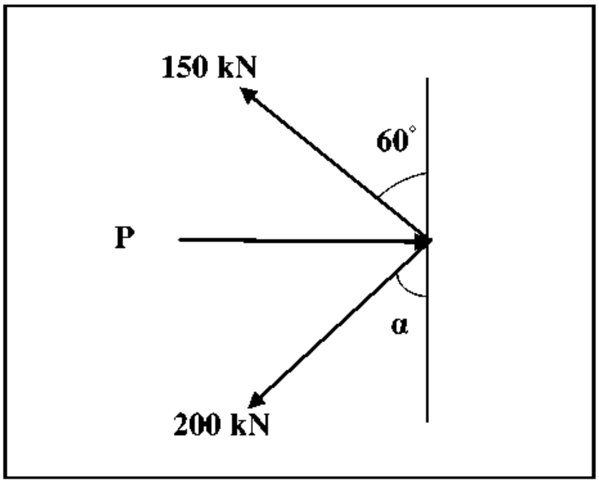
Ans. W= 18.25kN

6. A system of connected flexible cables, shown in fig is supporting two vertical forces 200 N and 250 N at points A and B. Determine the forces in various segments of the cable.

(Ans: T1= 224.14 N, T2=183.01 N, T3= 336.60 N, T4= 326.80 N)

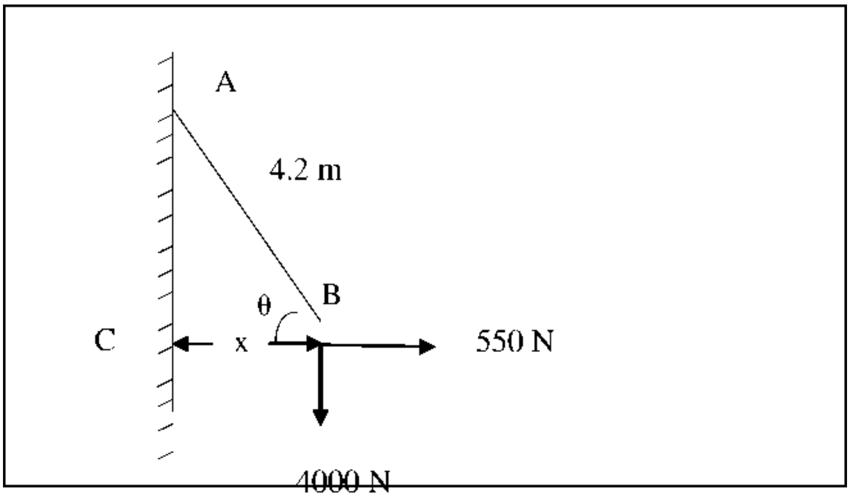


7. Determine the force P required to keep the system as shown in fig. in equilibrium.
   (Ans : P = 315.3 kN)



8. A body weighing 4000 N is suspended with a chain AB 4.2 m long. It is pulled by a horizontal force of 550 N as shown in figure. Find the force in the chain and the

lateral displacement (x) of the body.



Ans : θ= 82.16 , F = 4037.74 N , x = 0.57 m

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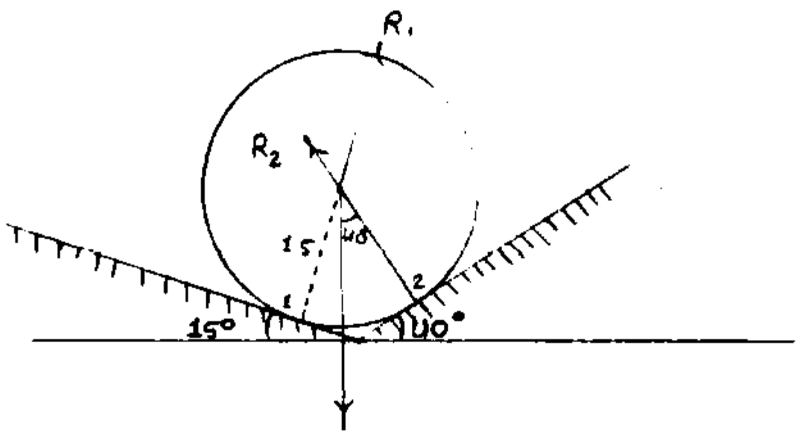
TYPE:3 EQULLIBRIUM CHECK

1. Two cylinders, each of diameter 40cm and mass 500N, rests in a rectangle channel of

base width 72cm.Find the reactions at all contact surfaces. Refer figure.

(Ans: 833.33N, 666.66N, 666.66N & 1000N)

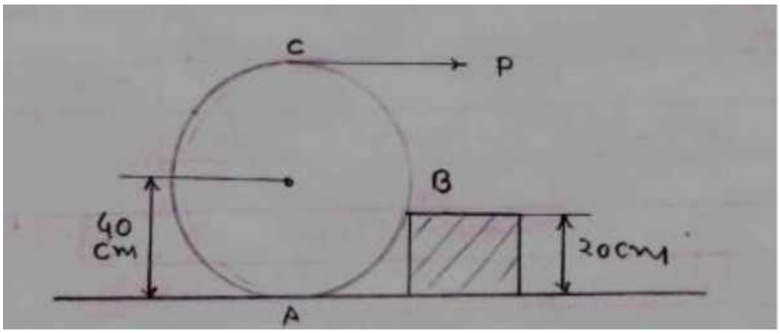
2. A smooth cylinder of radius 1.5 m is laying in a triangular groove one side of which Makes 15° angle and other side 40° angle with the horizontal. Find the reactions at the Surface of the contact if there is no friction and weight of cylinder is 10 kN. Refer Figure.



(Ans R1 = 7.84 kN, R2 = 3.15 kN)

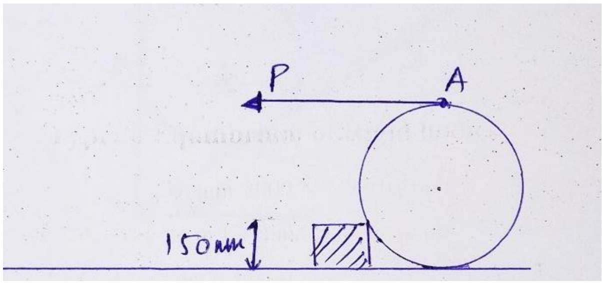
3. A roller of radius 40 cm weighing 3000 N is to be pulled over rectangular block of Height 20 cm in shown in figure by a horizontal force applied at the end of string Wound round the roller over the corner of the rectangular block. Also determine the

Magnitude and direction of reaction at A and B. All surfaces may be taken as smooth.



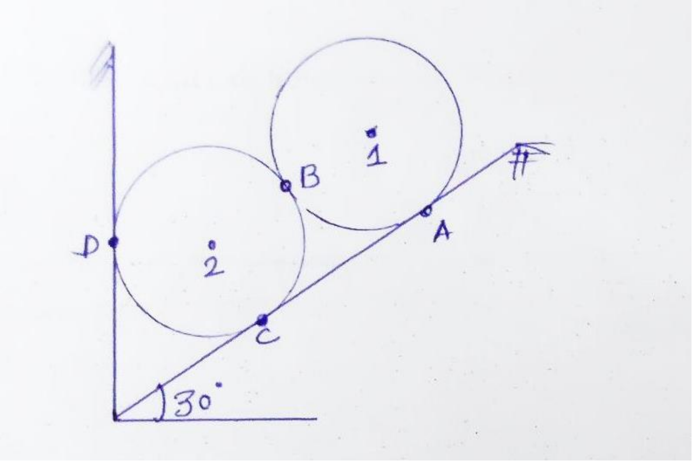
[ANS: (P=1732.05 N and R=3464.10N)]

4. A roller of radius 300 mm and weight 2000 N is to be pulled over a curb of height 150 mm by a horizontal force p shown in fig. Find i) the magnitude of force P required to just move the roller over the curb, ii) the least value of force P to be applied at A.



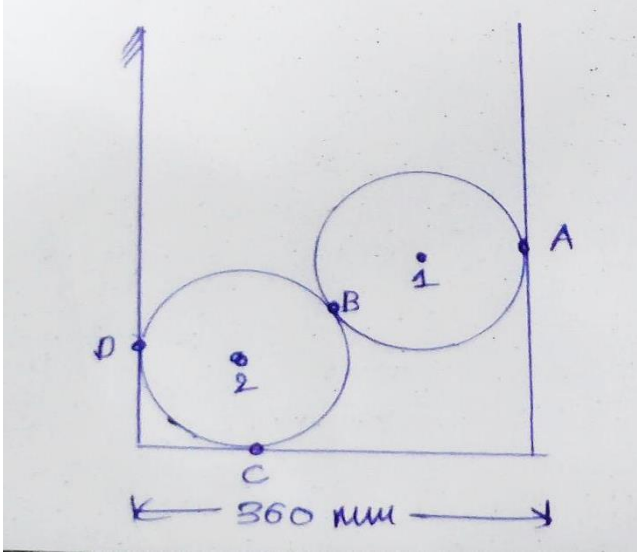
(Ans: P=1154.70 N, R_c=2309.40 N, P_{least}=1000 N)

5. Two identical rollers, each weighing 100 N are supported by an inclined plane and a vertical wall as a shown in fig. Assuming all the surfaces to be perfectly smooth, find the reaction induced at the points of supports A, B, C and D.
   (Ans: R_A=86.60 N , R_B=50 N, R_C=144.34 N and R_D=115.47 N)



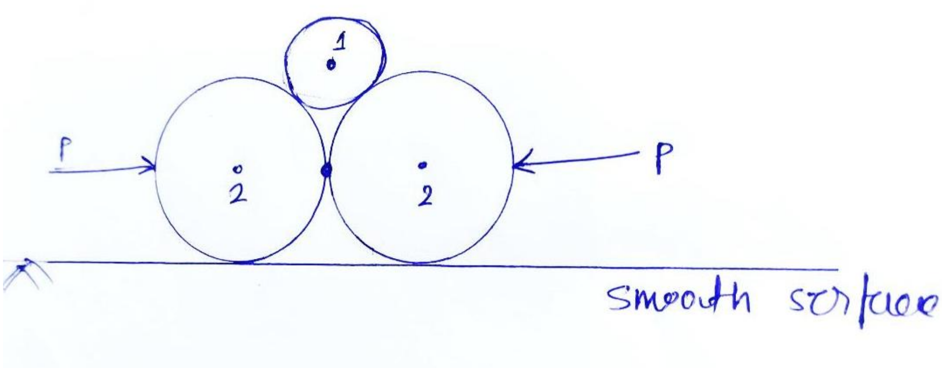
6. Two smooth sphere of radius 100 mm and weight 100 N rest in a channel as a shown in fig. Find the reaction at the point of contacts A, B, C and D.

(Ans: R_A=133.33 N , R_B=166.60 N, R_C=200 N and R_D=133.33 N)



7. Find the minimum (least) value of force P to keep the sphere in the position as a shown in fig. The radii of sphere-1 is 5 cm and sphere-2 is 10 cm. The weight of sphere-1 is 100 N and sphere-2 is 200 N.

(Ans: R1=250 N, R267.07 N, P=44.7 N)


1
P
2
2
P
smooth surface

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TYPE: 4 RESULTANT OF COPLANAR CONCURRENT FORCE SYSTEM IN SPACE

CH: 3- FRICTION

I) Ladder Friction

1) A uniform ladder, of length 5m, is supported by a horizontal floor at 'A' and a vertical wall at 'B' and makes an angle of 60° with the horizontal. Find the maximum distance 'x' up the ladder at which a man of weight 700N can stand without causing slipping of the ladder. The coefficient of friction between floor & ladder and wall & ladder is0.3. Neglect the weight of ladder.

   (Ans: Rw = 192.67 N, Rf= 642.17 N, x= 2.796 m)

2) A ladder 6 m long has a mass of 18 kg and its center of gravity is 2.4 m from the bottom. The ladder is placed against a vertical wall so that it makes an angle of 60° with the ground. How far up the ladder can a 72-kg man climb before the ladder is on the verge of slipping? The angle of friction at all contact surfaces is 15°.

   (Hint: μ=tan15° Ans: *Rw*=22.5 kg, *Fw*=6.03 kg, x = 3.15 m)

3) A ladder 7 m long rests against a vertical wall with which it makes an angle of 45° and resting one floor. If a man whose weight is one half of that the ladder, climbs it. At what distance along the ladder will he be when ladder is about to slip? M= 1/3 at wall and ½ at floor.

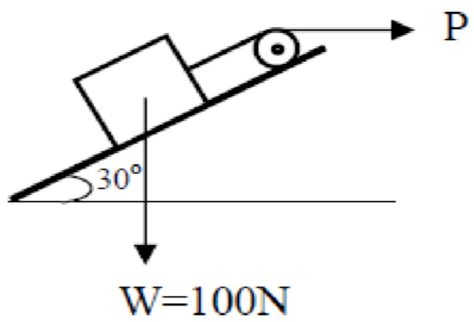
   (Ans: x= 5 m)

4) Define and explain: (i) Coefficient of friction (ii) Angle of friction (iii) Laws of dry friction.

II) Block Friction

5) Equilibrium of block is maintained by a pull *P* as shown in Fig. The coefficient of friction between block and surface is 0.2. Determine thevalues of *P* for which the block remains in equilibrium.

   (Ans: P= 87.88kN)

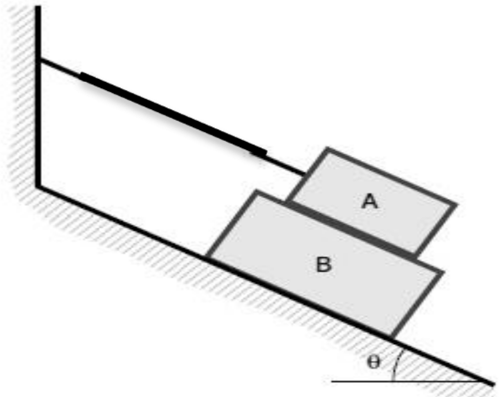


6) A 400 N block is resting on a rough horizontal surface for which the coefficient of friction is 0.40. Determine the force P required to cause motion to impend if applied to the block (a) horizontally or (b) downward at 30° with the horizontal.

(Ans: (a)160N (b)240.23N)

7) Block A in Fig. weighs 120 kN, block B weighs 200 kN, and the cord is parallel to the incline. If the coefficient of friction for all surfaces in contact is 0.25, determine the angle θ of the incline of which motion of B impends.

(Ans: θ=28.81°)

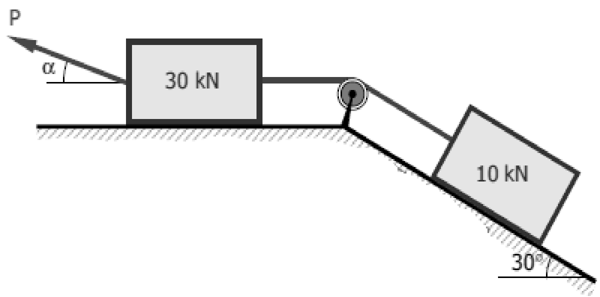


8) Referring example, no-7 if the coefficient of friction is 0.60 and θ = 30°, what force P applied to B acting down and parallel to the incline will start motion? What is the tension in the cord attached to A?

(Ans: T=122.35 kN)

9) Find the least value of P required to cause the system of blocks shown in Fig. to have impending motion to the left. The coefficient of friction under each block is 0.20.

(Ans: 12.5kN)



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CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Technology and Engineering
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CH: 5- CENTROID AND CENTRE OF GRAVITY

1) Write the formula for the centre of gravity of the Composite linear elements and Composite lamina.
2) Derive the formula of centre of gravity of a line or a rod .
3) Derive the formula of centre of gravity of quarter circle arc.
4) Derive the formula of centre of gravity of rectangular lamina.
5) Derive the formula of centre of gravity of quarter circle lamina.
6) Derive the formula of centre of gravity of triangle having base b and height h.
7) Find the CG of the thin homogeneous wire as shown in figure 1.

   (Ans: X= 16.67 cm & Y= 5.83cm)
8) Find the CG of the thin homogeneous wire as shown in figure 2.

   (Ans: X= 6.207 cm & Y= 6.519 cm)
9) Locate the CG of bar model as shown in figure 3.

   (Ans: X= 24.21 mm & Y= 101.57 mm)
10) Locate the centoid of a wire as shown in figure 4.

    (Ans: X= 32.19 cm & Y= 83.18 cm)
11) Locate the centroid of a lamina as shown in figure 5.

    (Ans: X=126.20 mm & Y= 42.96 mm)
12) Locate the centroid of a lamina as shown in figure 6.

    (Ans: X=20 mm & Y= 50.03 mm)
13) An unsymmetrical I section has the following dimensions in mm.

    Bottom flange – 250 x 100

    Top flange – 100 x 50

    Web – 300 x 50

    Determine the position of CG of the section.

    (Ans: X= 125mm & Y= 158.33mm)
14) Find the centroid of the lamina shown in figure 7.

    (Ans: X= 2.57 cm & Y= 4.81 cm)

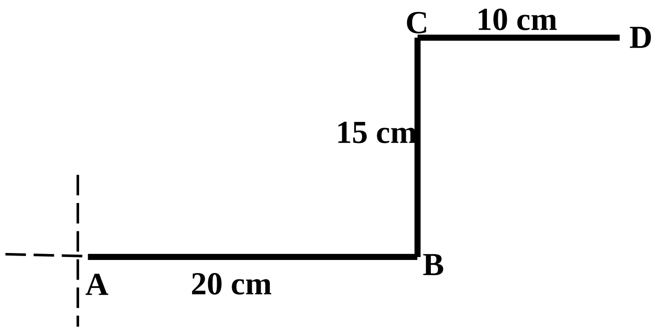

Figure 1

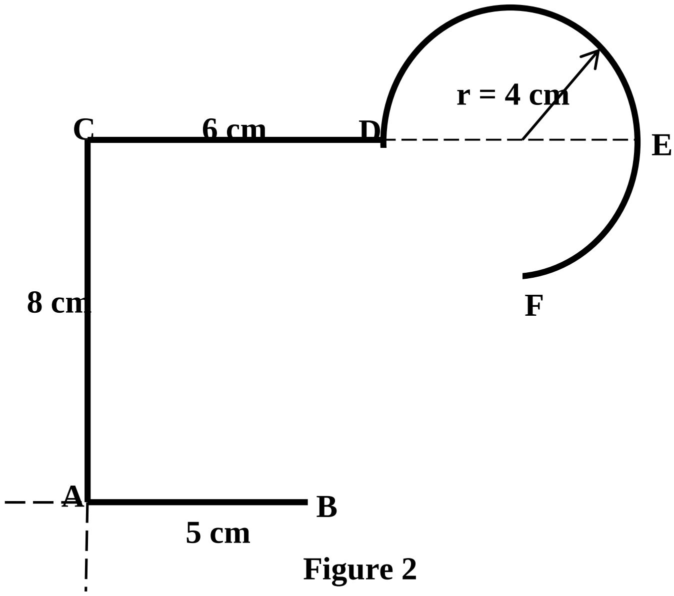

Figure 2

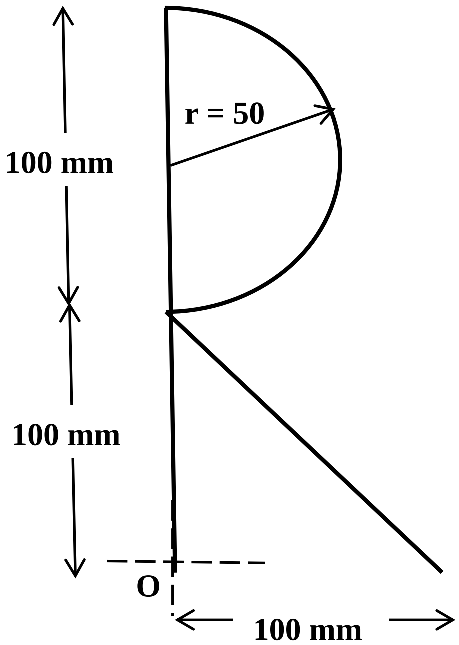

Figure 3

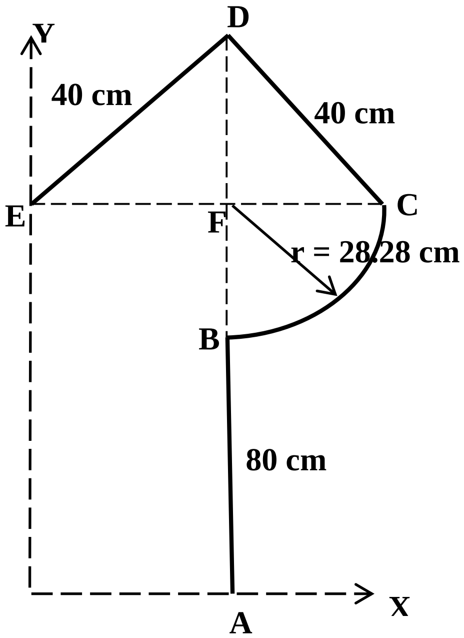

Figure 4

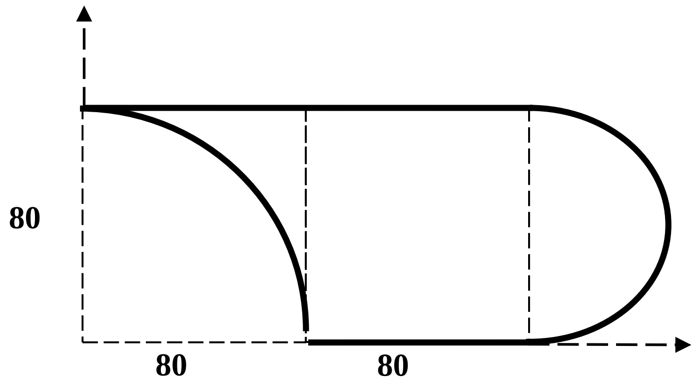

Figure 5

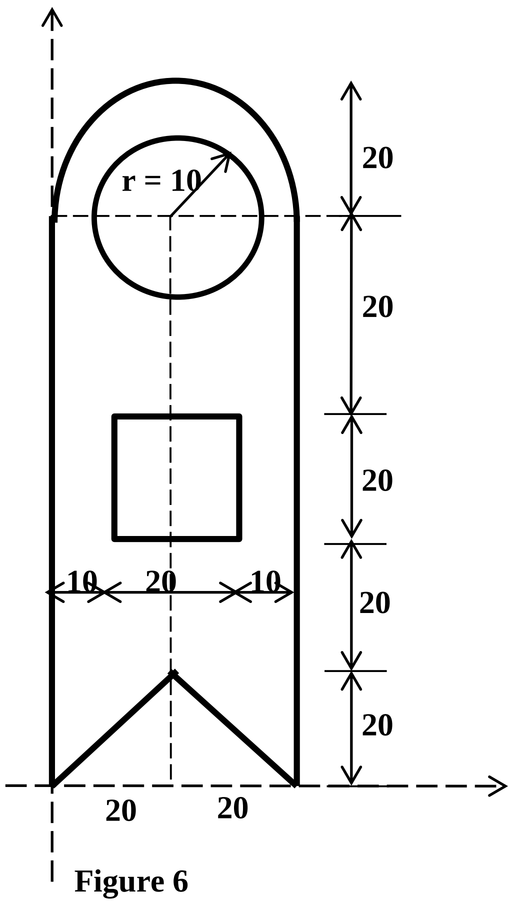

Figure 6

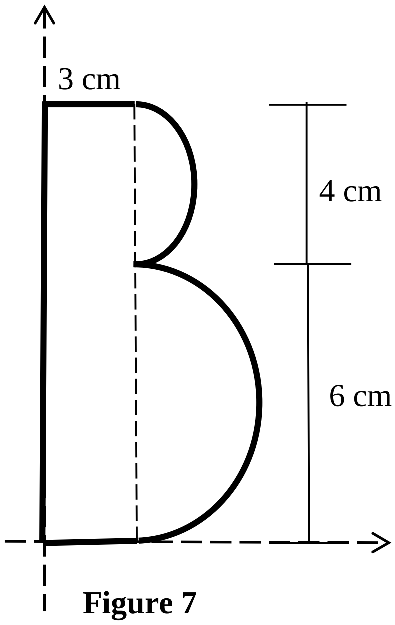

Figure 7

All dimensions are in cm

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